Numerical Modelling of Waves at Bretagne Sud Site

Luca Knab

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1 Motivation

The global energy transition necessitates a rapid expansion of renewable capacity, with offshore wind playing a pivotal role. The International Energy Agency (IEA) estimates that the technical potential for offshore wind exceeds 420,000 TWh per year—more than 18 times the current global electricity demand. Crucially, approximately 80% of these resources are located in waters deeper than 60 meters, where traditional fixed-bottom foundations are technically or economically unfeasible [1]. Consequently, the development of floating offshore wind turbines (FOWT) is essential to access high-energy deep-water sites.

However, the viability of floating structures in dynamic marine environments relies on a rigorous characterization of hydrodynamic loads. Design limits are governed not merely by deep-water statistics, but by the complex transformation of waves, via refraction, shoaling, and diffraction as they propagate across the continental shelf. This report addresses these challenges by performing a multi-scale analysis that integrates satellite observation, geometric-optics ray tracing, and Froude scaled experimental modelling to define precise, site-specific design conditions and structural loads.

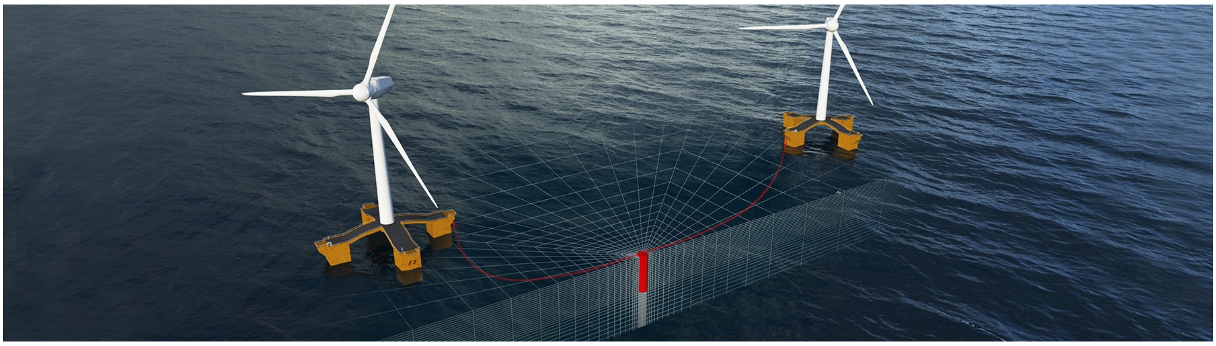


Figure 1: Illustrative rendering of a FOWT [2]

Part I
Satellite Imagery Analysis

2 Data Selection

The satellite-based analysis uses a Sentinel-2 Level-2A image acquired on 25 March 2020, accessed through the Copernicus Data Space Ecosystem Browser.[1] The study area is centred on the updated Bretagne Sud 1 hind-cast point at (47.3236°N, 3.5522°W), corresponding to ResourceCode grid node 117231, located about 1.41 km from the nominal project coordinates [3]. The scene encompasses the outer continental shelf, capturing the shoreward propagation of waves modeled in the subsequent ray-tracing simulations. The date was selected specifically for its minimal cloud cover, which ensures the wave crests are clearly visible.

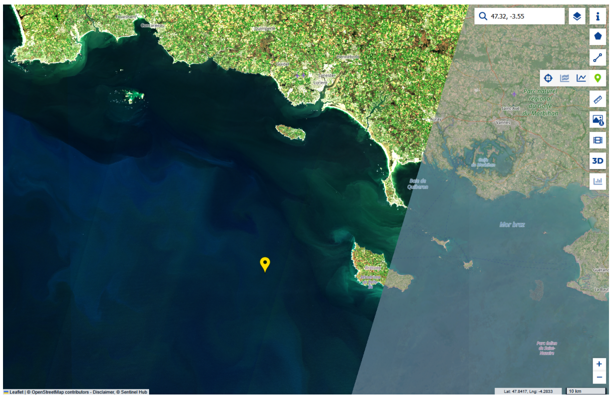


Figure 2: Sentinel-2 image of the Bretagne Sud region on 25 March 2020, viewed in the Copernicus Data Space Browser. The yellow marker indicates the Bretagne Sud 1 hind-cast point at 47.3236°N, 3.5522°W

For 25 March 2020, the ResourceCode hind-cast provides 24 hourly sea states at node 117231. The daily means of the sea states have been computed for comparison with observed values from the satelitte imageries.

Table 1: hind-cast sea-state parameters at ResourceCode node on 25 March 2020.

Parameter	Mean	Std. Dev.
Significant wave height H_s [m]	1.95	± 0.15
Mean energy period T_{02} [s]	7.28	± 0.90
Peak period T_p [s]	13.40	± 0.71
Coming-from direction θ_{RC} [°]	286.73	± 2.24
Water depth h [m]	93.31	± 1.29

Using the mean peak period, a reference deep-water wavelength is

$$L_0 = \frac{gT_p^2}{2\pi} \qquad \Rightarrow \qquad \overline{L_0} \approx 281\,\text{m},$$

with an indicative spread $\sigma_{L_0} \approx 30\,\text{m}$ arising from the daily variability in T_p. Since $h/\overline{L_0} \approx 0.33$, the waves propagate in intermediate depth, so a more precise comparison is obtained from the full dispersion relation

$$\omega^2 = gk\tanh(kh), \qquad \omega = \frac{2\pi}{T_p},$$

which gives

$$L_{\text{theor}} \approx 272\,\text{m}.$$

[1] https://browser.dataspace.copernicus.eu/

To quantify its uncertainty, linear error propagation is applied using

$$\sigma_{L_{\text{theor}}} \approx \left| \frac{\partial L}{\partial T} \right|_{T=T_{\text{p}}} \sigma_{T_{\text{p}}},$$

where $\sigma_{T_{\text{p}}} = 0.71\,\text{s}$ is the daily standard deviation of the peak period. Evaluating $\partial L/\partial T$ numerically from the implicit dispersion relation yields

$$\sigma_{L_{\text{theor}}} \approx 22\,\text{m}.$$

Thus, the intermediate-water theoretical wavelength for the satellite overpass conditions is

$$L_{\text{theor}} = 272 \pm 22\,\text{m}, \qquad h/L_{\text{theor}} \approx 0.34,$$

which provides the benchmark for comparison with the crest spacings extracted from the Sentinel–2 image conducted in section 3.

3 Wave Properties from Satellite Imagery

The Sentinel–2 image allows extraction of two directly measurable quantities in the offshore region near Bretagne Sud 1, the dominant wavelength L_{obs} and the coming-from direction θ_{obs}. Both are estimated in the Copernicus browser at the native 10 m resolution, assuming that the visible crest pattern corresponds to the dominant spectral peak and that small-scale radiometric variations do not bias the measurements.

3.1 Observed Wavelength

Wavelength is obtained by measuring crest-to-crest spacings along a transect oriented perpendicular to the visible wave fronts. In the Copernicus browser, a polyline is drawn across several successive crests, and individual distances are read from the *Measure distance* tool.

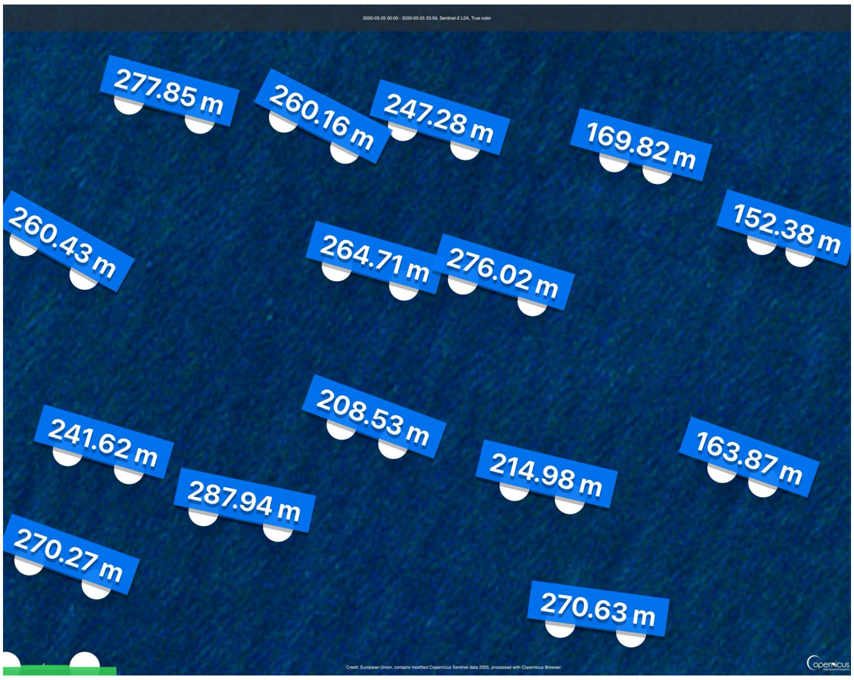


Figure 3: Sentinel–2 excerpt (25 March 2020) showing the transect used to measure crest spacings. Blue labels indicate individual crest-to-crest distances

Along the transect in Figure 3, fifteen crest spacings between approximately 150 and 290 m were recorded. Their sample statistics are:

$$L_{\text{obs}} \approx 238\,\text{m}, \qquad \sigma_L \approx 45\,\text{m},$$

so the dominant visible wavelength is about 238 m with $\sim 20\%$ variability along the sampled segment. The corresponding standard error of the mean is roughly 12 m, reflecting both natural modulation of the wave field and measurement uncertainty associated with the 10 m pixel size and manual crest selection.

To compare with the resourcecode dataset, the theoretical intermediate-water wavelength for the daily mean peak period is obtained by solving the dispersion relation as described in section 2. The observed wavelength is approximately 34 m (12%) shorter than the theoretical value (L_{theor}). This deviation is not statistically significant, as the measurement spread ($\sigma_L \approx 45\,\text{m}$) and hind-cast variability ($\sigma_{L_0} \approx 22\,\text{m}$) result in overlapping confidence intervals. Consequently, the satellite observations are broadly consistent with the hind-cast data, with residual discrepancies attributed to local bathymetric heterogeneity, spectral spreading, and the temporal resolution mismatch between instantaneous imagery and daily-mean statistics.

3.2 Observed Wave Propagation Direction

The incoming wave direction is estimated from the orientation of individual crests in the Sentinel–2 image. Crest segments are digitised in the Copernicus viewer, and their azimuths α are measured clockwise from geographic north. Assuming that crests are orthogonal to the direction of propagation, the coming-from direction follows as

$$\theta_{\text{obs}} = (90^\circ - \alpha) \bmod 360^\circ$$

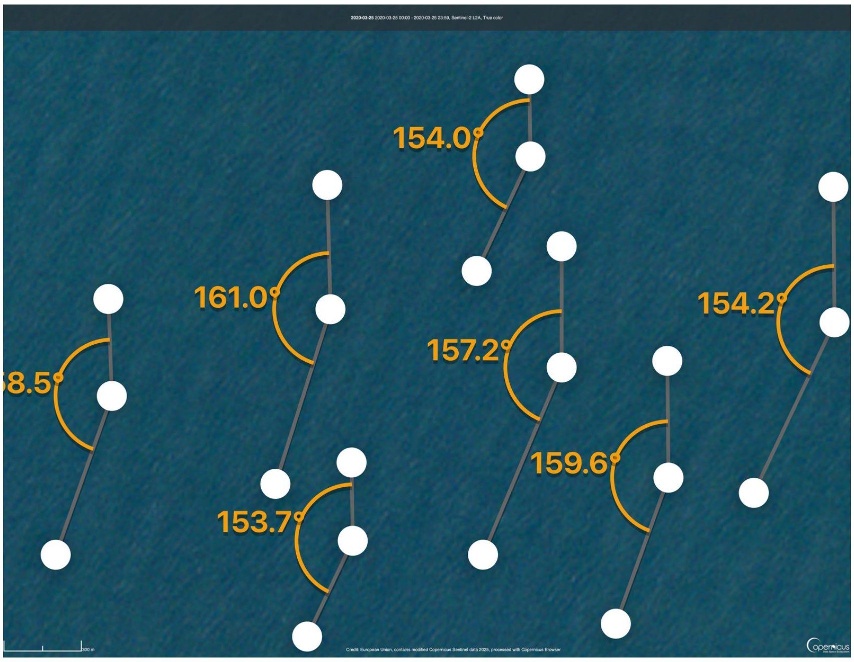


Figure 4: Crest orientation measurements extracted from the Sentinel–2 image on 25 March 2020. Grey line segments indicate digitised crest orientations; orange labels show the measured azimuths

Seven crest segments were measured, including the cropped value at the left of the frame. The resulting crest azimuths and converted coming-from directions are summarised in Table 2.

Table 2: Crest azimuths α (clockwise from north) and corresponding coming-from directions θ_{obs}.

	0	1	2	3	4	5	6
α (°)	158.5	161.0	153.7	154.0	157.2	159.6	154.2
θ_{obs} (°)	291.5	289.0	296.3	296.0	292.8	290.4	295.8
Mean	$\overline{\alpha} = 156.9°$, $\overline{\theta}_{\text{obs}} = 293.1°$						
Std. dev.	$\sigma_\alpha = 3.0°$, $\sigma_\theta = 3.0°$						

The crest orientations are tightly clustered around $\overline{\alpha} \approx 156.9°$ with a spread of $\pm 3°$, consistent with a coherent swell system. The corresponding propagation directions yield

$$\overline{\theta}_{\text{obs}} \approx 293.10°, \qquad \sigma_\theta \approx 3.00°.$$

This indicates waves arriving from the west–north-west, with only minor directional variability across the sampled region.

For comparison, the hind-cast for 25 March 2020 gives

$$\theta_{\text{RC}} = 286.73° \pm 2.24°.$$

The satellite estimate is therefore about 6.40° more northerly than the daily hind-cast mean. This difference lies within the combined uncertainty ranges and may reflect moderate refraction between the hind-cast node and the measurement area, sub-daily directional variability, or contributions from secondary wave components that are more visible in the image than in bulk spectral statistics.

3.3 Qualitative Assessment of Refraction

To complement the quantitative analysis in subsection 3.1 and subsection 3.2, a qualitative inspection of the coastal wave field was conducted to identify local propagation phenomena. Figure 5 presents a contrast-enhanced Sentinel-2 subscene of the northern shoreline of Belle-Île-en-Mer, illustrating the evolution of the incident swell over changing bathymetry.

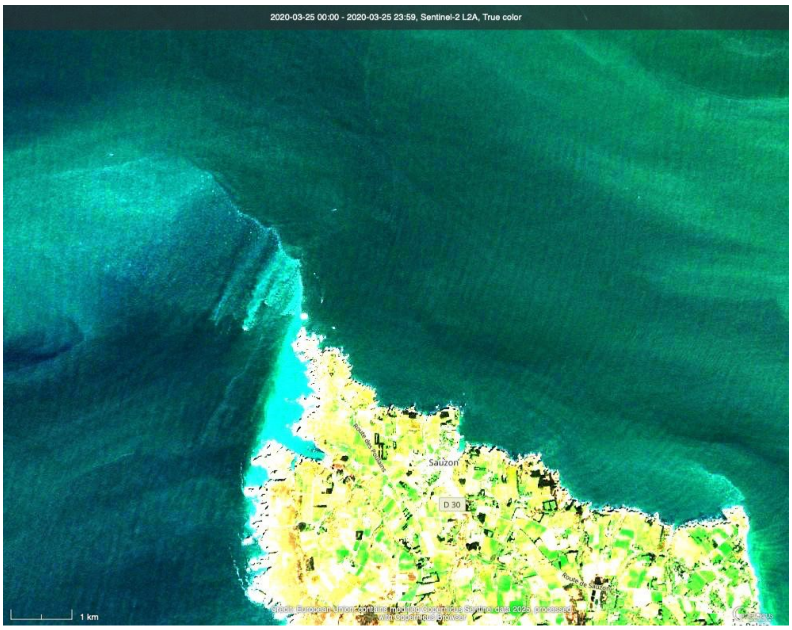


Figure 5: Sentinel–2 image on 25 March 2020 showing wave patterns near the northern coast of Belle–Île–en–Mer. Offshore crests are broadly parallel; nearshore, they bend around the headland, tighten in spacing, and weaken in the lee

Offshore of the island, the crests appear long, continuous, and only gently curved, consistent with the organised swell identified along the offshore transect. Moving landward, however, the crests bend progressively toward the shoreline, seen in the bottom left corner, become more closely spaced, and eventually diminish in contrast in the lee of the headland. This transition reflects classical bathymetric refraction: as water depth decreases, the local phase speed reduces, causing wave rays to rotate toward the normal of the depth contours and to converge along the north-facing coast. The weakening of the signal immediately behind the headland further indicates diffraction into a geometric shadow zone [4].

3.4 Discussion

The satellite imagery analysis provides an independent, semi-quantitative validation of the ResourceCode hind-cast dataset for the Bretagne Sud site. The quantitative comparison reveals general consistency between the remote sensing data and the numerical model. The observed wavelength is approximately 34 m (12%) shorter than the theoretical value derived from the hind-cast. This deviation is not statistically significant, as the measurement spread ($\sigma_L \approx 45\,\mathrm{m}$) and hind-cast variability ($\sigma_{L_0} \approx 22\,\mathrm{m}$) result in overlapping confidence intervals. Similarly, the directional analysis confirms a coherent WNW swell system. The observed northerly offset of approximately 6° relative to the hind-cast mean lies within the combined uncertainty ranges and likely reflects moderate refraction occurring between the deep-water node and the measurement transect.

The precision of these measurements is constrained by several limiting factors. The primary source of uncertainty is the temporal discrepancy between the datasets; the hind-cast parameters are daily averages which smooth out sub-daily fluctuations in wind forcing and tidal currents, whereas the satellite image captures a single, instantaneous realisation of the sea state. Consequently, exact numerical alignment is not expected. Furthermore, measurement uncertainty plays a non-negligible role. The Sentinel-2 spatial resolution of 10 m introduces a quantization error of roughly 4% for wavelengths of this magnitude, and the manual identification of wave crests is subject to interpretation, particularly in areas of lower contrast. The dataset is also spatially constrained and small (N=15 for wavelength, N=7 for direction), which limits the ability to fully characterize the spectral width or the directional spreading of the sea state.

Regarding the nearshore analysis, a quantitative reconstruction of crest turning angles and local wavelength changes would require systematic measurements at multiple depths and are left as future work. In the present analysis, Figure 5 is used qualitatively to demonstrate that the regional bathymetry produces strong focusing and shadowing in the same general sector as the Bretagne Sud 1 site. This qualitative confirmation of bathymetric control validates the physical basis for the variability observed in the quantitative analysis.

Part II
Ray Tracing

4 Input Wave Conditions

4.1 Methodology

The offshore forcing for the ray-tracing simulations is defined at the ResourceCode node closest to the western model boundary (Figure 6). This ensures that the initialized rays represent waves entering the domain directly from deep water and that the subsequent transformation reflects the true bathymetric control of the shelf.

Extreme conditions at this location were obtained from the multivariate Principal Component Analysis (PCA) of the joint (H_s, T_p) distribution performed in the previous report. PCA preserves the physical correlation between wave height and period. The resulting contour points therefore represent physically consistent, upper-bound offshore conditions appropriate for refraction-sensitive modelling.

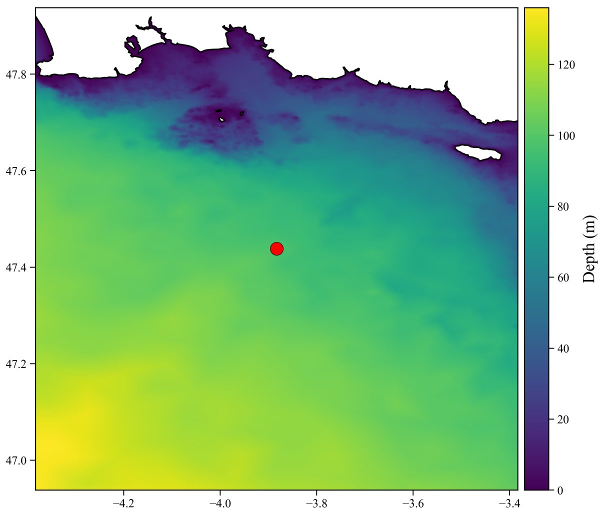


Figure 6: Location of the selected ResourceCode boundary node (47.438°N, −3.822°W) used for ray initialization

Directional information was derived from the POT analysis, yielding a robust mean coming-from direction of 260.4° (WSW), which was converted to a going-to heading of 80.4° for ray propagation. Three scenarios were defined to explore the period dependence of refraction: a baseline case representing a typical North Atlantic storm ($T = 14$ s)—derived from the central tendency of extreme events identified in the prior POT and BM analyses—and two PCA-derived design scenarios for the 1-year and 50-year return periods (Table 3).

4.2 Results

The PCA-based design states yield long peak periods compared to empirical deep-water estimates. For the 1-year return period,

$$H_{\mathrm{s,1y}} = 12.55\,\mathrm{m}, \qquad T_{\mathrm{p,1y}} = 19.74\,\mathrm{s},$$

and for the 50-year return period,

$$H_{\mathrm{s,50y}} = 18.94\,\mathrm{m}, \qquad T_{\mathrm{p,50y}} = 22.76\,\mathrm{s}.$$

A deep-water steepness check,

$$T_{\mathrm{check}} \approx 12.7\sqrt{\frac{H_{\mathrm{s}}}{g}},$$

yields $T_{\mathrm{check}} \approx 14.4\,\mathrm{s}$ (1-year) and $17.7\,\mathrm{s}$ (50-year) [5]. The PCA periods therefore exceed both T_{check} and typical T_p values (12–18 s) derived from univariate fits. This behaviour is expected:

selecting the uppermost point on each environmental contour forces large H_s to pair with the long-period tail of the observed joint distribution. These long periods reflect a conservative but physically consistent representation of extreme swell, ensuring that the ray-tracing captures the maximum plausible degree of refraction over the shelf.

These states form the offshore wave boundary conditions for all subsequent ray-tracing and local wave-height estimation.

Table 3: Offshore design sea states defined at the boundary node (47.438°N, −3.822°W).

Scenario	$H_{s,off}$ (m)	T_p (s)	Direction (° going-to)
Baseline	-	14.00	80.4
1-Year	12.55	19.74	80.4
50-Year	18.94	22.76	80.4

5 Model Domain

5.1 Methodology

Bathymetric data was sourced from the Global Multi-Resolution Topography (GMRT) grid, provided as a 1225 × 703 ASCII raster in WGS84 coordinates [6]. The spatial resolution is approximately 0.0011°, corresponding to roughly 83 m (East-West) by 122 m (North-South). A local Cartesian coordinate system (x, y) was established relative to the south-west grid corner using an Earth-radius approximation. The Bretagne Sud 1 hind-cast point was projected onto this grid to determine the precise local coordinates and water depth for the subsequent analysis.

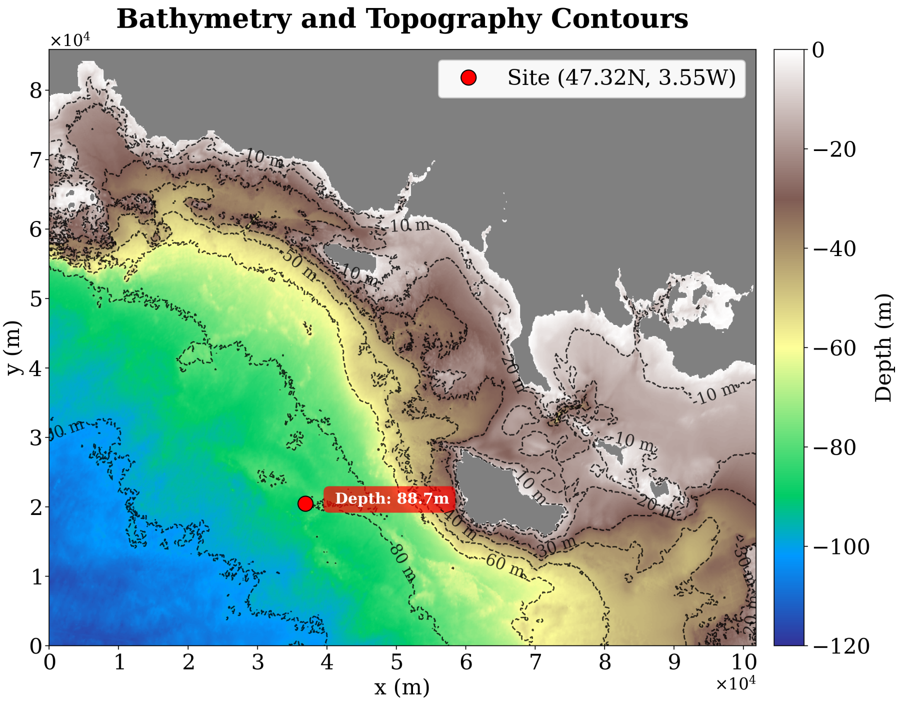


Figure 7: Bathymetry and topography of the Bretagne Sud model domain. Contours show depth in 10–20 m intervals. The red marker denotes the project site, located on the mid-shelf slope in approximately 88.7 m water depth

5.2 Results

The bathymetry shown in Figure 7 is characterized by a gently sloping continental shelf deepening from the shallow nearshore ($h \lesssim 20\,\mathrm{m}$) to depths exceeding 100 m in the southwest. The regional slope is punctuated by complex local topography. In the northeast, the archipelago exhibits high steepness, evidenced by the closely spaced contours surrounding the small islands.

South of Belle-Île-en-Mer, the landward indentation of the 30 m contour indicates a local submarine depression or valley extending onto the shelf. From a wave mechanics perspective, deeper channels generally induce divergence along the axis; however, the bathymetric gradients along the feature's margins may simultaneously generate refractive scattering or focusing on adjacent banks [7]. The project site is situated on the mid-shelf slope at a depth of 88.7 m, where the incident wave field is likely modulated by the integration of these upstream refractive effects.

6 Model Configuration

6.1 Methodology

The ray-tracing simulations employ a geometric-optics model based on the linear dispersion relation:

$$\omega^2 = gk \tanh(kh)$$

Wave propagation is governed by the eikonal equation, where local depth variations modulate group velocity, driving refraction. Energy conservation along rays allows ray density to proxy for relative wave energy flux [8].

Simulations were performed on the raw, unsmoothed GMRT bathymetry grid. An initial test with Gaussian smoothing ($\sigma = 2$ cells) was discarded because it suppressed steep gradients critical for refraction. The simulation duration was set to $T_{\mathrm{sim}} = 9\,000\,\mathrm{s}$ (2.5 h). The time discretization utilized $n_t = 20,000$ steps, yielding a computation time step of:

$$\Delta t = \frac{T_{\mathrm{sim}}}{n_t} = 0.45\,\mathrm{s}$$

This ensures a temporal resolution of at least 30 points per period for the shortest design wave ($T = 14$ s). For each scenario (14 s, 1-year, 50-year), two ensembles were computed: a sparse set (100 rays) for trajectory visualization and a dense set (1,000 rays) for energy flux estimation.

6.2 Results

The final numerical configuration successfully integrates the high-resolution $1\,225 \times 703$ depth field without numerical instability for the majority of rays. The selected time step and integration duration proved sufficient to track wave propagation from the deep-water boundary to the coastline across all tested periods.

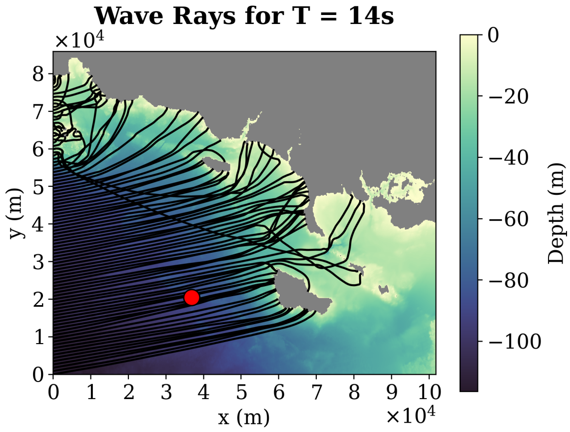
6.3 Discussion

The decision to use the raw GMRT bathymetry prioritizes physical accuracy over numerical smoothness. While this preserves essential refractive features like narrow channels and steep banks, it introduces a trade-off where a small minority of rays initialized on sharp gradients may exhibit pathological behavior (looping). These artefacts are localized and do not compromise the global energy flux estimates derived from the dense ray ensembles. The configuration neglects higher-order effects such as current refraction and bottom friction.

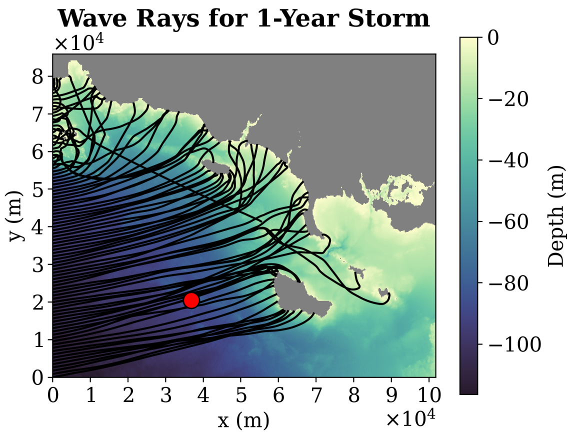
7 Ray-Tracing Runs and Quality Checks

7.1 Methodology

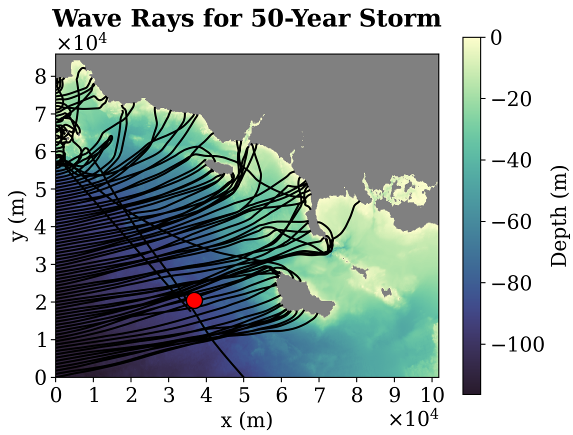
Rays were initialized along the western boundary of the computational domain with uniform latitudinal spacing. The initial propagation direction was fixed at 80.4° (going-to), corresponding to the dominant WSW storm sector (260.4° coming-from) identified in the offshore analysis. As described in section 4, three wave periods were simulated to assess frequency-dependent propagation.



(a) $T_{\text{ray}} = 14\,\text{s}$



(b) $T_{\text{ray}} = 19.74\,\text{s}$ (1-Year)



(c) $T_{\text{ray}} = 22.76\,\text{s}$ (50-Year)

Figure 8: Ray trajectories for the three simulated wave periods. Black curves show ray paths over the GMRT bathymetry; the red marker denotes the project site

7.2 Results

The trajectory visualizations (Figure 8) reveal distinct propagation behaviors. For the 14s baseline storm (Figure 8a), rays entering from the western boundary between $y = 0$ and $y \approx 80,000\,\text{m}$ exhibit a largely uniform trajectory across the domain. Irregularities are confined to the north, where rays interact with the archipelago. Observable bending is restricted to the mid-shelf ridge and the immediate vicinity of coastal headlands. Consequently, the project site lies within a wide corridor of rays characterized by relatively constant spacing and direction.

This diffuse pattern reflects the intermediate depth regime at the boundary ($h/L_0 \approx 0.26-0.39$), where bottom control is insufficient to drive strong convergence.

In contrast to the baseline, the 1-year and 50-year design states (Figure 8b and Figure 8c) exhibit markedly stronger refraction. As the period increases, the deep-water wavelength grows ($L_0 \approx 510\,\mathrm{m}$ and $880\,\mathrm{m}$, respectively), significantly reducing the depth ratio to $h/L_0 \approx 0.12-0.18$ at the boundary. Consequently, rays begin refracting further offshore and wrap tightly around the mid-shelf ridge and the shoals southwest of the site.

This mechanism generates a dense bundle of converging rays that directly intersects the project site. The 50-year scenario ($T = 22.76$ s) displays the most pronounced focusing, confirming that the regional bathymetry acts as a lens for extreme long-period swell. Across these simulations, a small number of rays initialized over steep nearshore gradients in the north-east archipelago exhibit anomalous looping trajectories. These are identified as numerical artifacts inherent to the use of unsmoothed bathymetry and do not reflect physical wave propagation [4].

7.3 Discussion

The simulations demonstrate a clear period-dependence in the wave climate at the project site. As the wave period increases from 14 s to the design extremes, the propagation regime shifts from weak refraction to strong bathymetric control. The mid-shelf ridge acts as a lens for long-period swell, systematically focusing energy toward the site. This confirms that the local wave height amplification is a function of the specific frequency content of extreme storms, validating the use of the coupled PCA-ray tracing approach. The anomalous rays observed in the 50-year simulation highlight the sensitivity of the geometric optics approximation to grid roughness but are localized enough to not compromise the validity of the global focusing pattern.

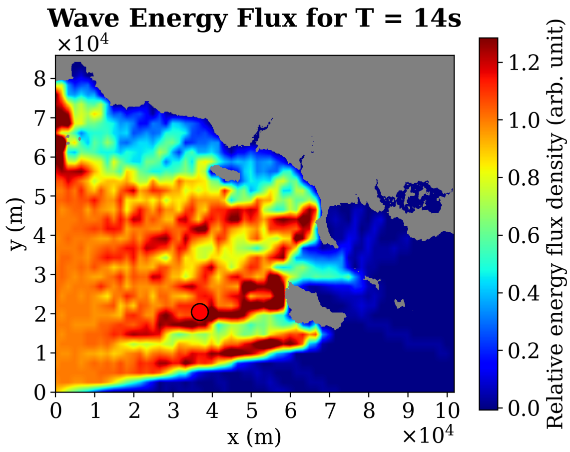
8 Wave Energy Flux Distribution

8.1 Methodology

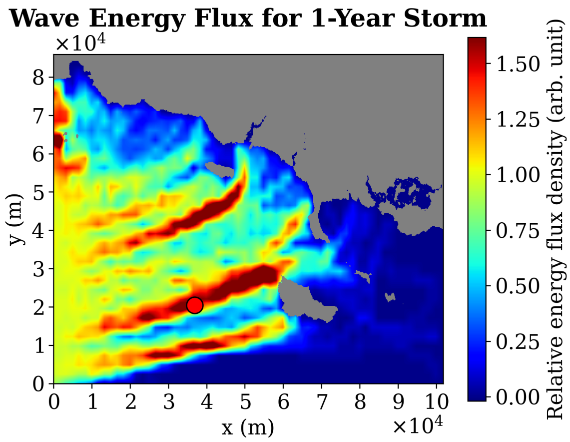
The 1,000-ray simulations are used to quantify the spatial redistribution of wave energy due to refraction. Ray positions were sampled over the integration duration and mapped onto a regular (x, y)grid . A ray-density field was computed by counting ray passages per cell, which serves as a proxy for the depth-integrated energy flux $F = Ec_\mathrm{g}$ (where c_g is group velocity). Assuming energy conservation along ray tubes, this field is normalized by the offshore boundary value to derive the relative energy flux density factor:

$$\frac{E}{E_0} = \frac{F}{F_0}$$

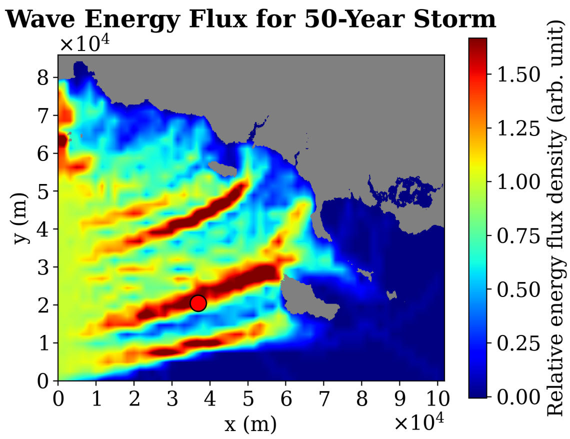
This dimensionless factor allows the amplification effects of bathymetry to be applied to any offshore significant wave height H_s [8].



(a) $T_{\text{ray}} = 14\,\text{s}$



(b) $T_{\text{ray}} = 19.74\,\text{s}$ (1-Year)



(c) $T_{\text{ray}} = 22.76\,\text{s}$ (50-Year)

Figure 9: Relative wave energy flux density (E/E_0) for the baseline (top) and extreme design states (bottom). Warmer colors indicate focusing (amplification), cooler colors indicate shadowing. The red marker denotes the project site

The flux density maps (Figure 9) quantify the focusing observed in the trajectory analysis. For the 14 s baseline storm (Figure 9a), energy is distributed relatively broadly across the mid-shelf. The project site experiences mild amplification, with E/E_0 values ranging between 0.7 and 1.1. Shadow zones behind islands remain weak ($E/E_0 > 0.4$), indicating diffuse propagation.

In contrast, the 1-year and 50-year design states (Figure 9b and Figure 9c) exhibit sharp energy gradients. Refraction organizes the wave field into distinct high-flux bands ("caustics") aligned with the mid-shelf ridge. Within these bands, the relative energy factor reaches peak intensities of 1.4–1.5. The project site is systematically located within one of these primary focusing corridors. Specifically, for the 1-year storm ($T = 19.74$ s), the local amplification factor is approximately 1.3. For the 50-year storm ($T = 22.76$ s), the focusing intensifies, yielding a factor of approximately 1.4. Conversely, adjacent sheltered areas experience significant energy deficits, with E/E_0 dropping below 0.3.

8.2 Discussion

The analysis demonstrates that the energy landscape at Bretagne Sud 1 is highly sensitive to wave period. While short-period storms result in a near-average energy distribution, extreme long-period swell is subject to strong bathymetric lensing. This systematic concentration of energy is driven by the interaction between the mid-shelf ridge and the adjacent submarine valley.

Physically, the deep valley discussed in section 5 induces higher phase speeds, causing wavefronts to diverge away from the channel axis. Simultaneously, the shallower mid-shelf ridge slows the waves, causing convergence. The sharp contrast in phase speeds between these adjacent features steers the energy out of the valley and focuses it onto the ridge flank where the project site is located. This mechanism results in local amplification factors of 30–40% relative to the offshore boundary, confirming that the site is located in a bathymetric "hotspot" for westerly swell.

9 Wave Height Estimates at the Project Site

9.1 Methodology

The relative energy factors at the project site are used to translate offshore contour heights into local significant wave heights. Under linear theory, the depth-integrated energy flux satisfies

$$F = E c_{\mathrm{g}} \propto H_{\mathrm{s}}^2 c_{\mathrm{g}},$$

where H_{s} is the significant wave height. Between the western boundary (depth about 100–120 m) and the project site ($h \approx 88.7\,\mathrm{m}$), changes in group velocity are small for the long periods considered, so the local height can be approximated as

$$\frac{H_{\mathrm{s,loc}}}{H_{\mathrm{s,off}}} \approx \sqrt{\frac{E}{E_0}},$$

where $H_{\mathrm{s,off}}$ is the offshore contour value and E/E_0 the ray-based energy factor at the site.

For the 1-year case, the energy map in Figure 9b gives $E/E_0 \approx 1.3$ at the project location. For the 50-year case, Figure 9c indicates a slightly stronger focusing with $E/E_0 \approx 1.4$. These values are combined with the offshore heights in Table 3 to obtain local $H_{\mathrm{s,loc}}$. Linear dispersion at $h = 88.7\,\mathrm{m}$ is then used to estimate the associated wavelengths as a consistency check [9].

9.2 Results

For the 1-year design storm, the offshore contour gives $H_{\mathrm{s,off}} = 12.55\,\mathrm{m}$ and $T_{\mathrm{p}} = 19.74\,\mathrm{s}$. With $E/E_0 \approx 1.3$,

$$H_{\mathrm{s,loc}} \approx H_{\mathrm{s,off}} \sqrt{\frac{E}{E_0}} \approx 12.55\sqrt{1.3}\,\mathrm{m} \approx 14.3\,\mathrm{m}.$$

Solving the dispersion relation at $h = 88.7\,\mathrm{m}$ for $T_{\mathrm{p}} = 19.74\,\mathrm{s}$ gives a local wavelength $L \approx 500\,\mathrm{m}$, so that $h/L \approx 0.18$ and $H_{\mathrm{s,loc}}/L \approx 0.029$.

For the 50-year design storm, $H_{\mathrm{s,off}} = 18.94\,\mathrm{m}$ and $T_{\mathrm{p}} = 22.76\,\mathrm{s}$. With $E/E_0 \approx 1.4$,

$$H_{\mathrm{s,loc}} \approx 18.94\sqrt{1.4}\,\mathrm{m} \approx 22.4\,\mathrm{m}.$$

At the same depth, the dispersion relation for $T_{\mathrm{p}} = 22.76\,\mathrm{s}$ yields $L \approx 610\,\mathrm{m}$, so that $h/L \approx 0.15$ and $H_{\mathrm{s,loc}}/L \approx 0.037$.

9.3 Discussion

The ray-based amplification factors increase the local significant wave height at the project site by roughly 15–20 % relative to the offshore contour heights. At the same time, the local steepness $H_{s,loc}/L$ remains well below typical breaking thresholds and within the regime where linear wave theory and geometric-optics ray tracing remain applicable. Both design states lie in intermediate water with moderate steepness, so the linear-ray framework is appropriate for a first estimate of design loads.

These results confirm that the site experiences systematically higher wave heights than would be inferred from offshore statistics alone, owing to bathymetric focusing over the mid-shelf ridge.

10 Summary of Design Conditions at the Project Site

10.1 Methodology

To provide a compact set of inputs for the load calculations in Part III, the offshore environmental-contour states are combined with the ray-tracing amplification factors to define local design sea states at the project site. The location is fixed at the bathymetric grid point corresponding to $(47.32°\text{N}, 3.55°\text{W})$ with water depth $h = 88.7\,\text{m}$. For each return period, the local significant wave height $H_{s,loc}$ is taken from the ray-based estimates above, the peak period T_p from the PCA environmental contour, and the mean direction from the POT directional analysis.

10.2 Results

Table 4: Local design sea states at the project site obtained by combining offshore environmental-contour conditions with ray-tracing-derived energy amplification. Directions are "coming-from," measured clockwise from north.

Case	h (m)	$H_{s,loc}$ (m)	T_p (s)	Direction (°)	Sector
1-year	88.7	14.3	19.74	260.4	WSW
50-year	88.7	22.4	22.76	260.4	WSW

10.3 Discussion

Both design states correspond to long-period swell in intermediate water, with local heights amplified by refraction and focusing over the mid-shelf ridge and surrounding banks. The 50-year storm is characterised by a substantially larger local significant wave height and longer peak period than the 1-year case, as well as slightly stronger focusing at the site.

These local sea states form a consistent bridge between the offshore hind-cast-based extreme-value analysis and the monopile load calculations in Part III. They incorporate not only the joint statistics of H_s and T_p offshore but also the spatial transformation of the wave field over the bathymetry.

Part III

Wave Loads on a Vertical Cylinder

This section estimates the horizontal wave load on a full-scale monopile by upscaling laboratory measurements. The analysis enforces Froude similarity to ensure dynamic consistency between the model and prototype. The experimental results are subsequently validated against a theoretical benchmark derived from the MacCamy-Fuchs linear diffraction theory. This comparison aims to verify whether Froude scaling of laboratory data yields reliable order-of-magnitude estimates for inline forces under moderate sea states typical of the Part I analysis.

11 Methodology

11.1 Experimental Configuration

Laboratory tests were conducted in a wave flume with a constant depth of $h_m = 0.60$ m and a width of 0.50 m. A vertical cylinder was mounted on the centerline, 12.41 m from the wavemaker (Figure 10). The setup includes a sloping beach to minimize reflection, ensuring that the generated waves approximate a progressive wave field.

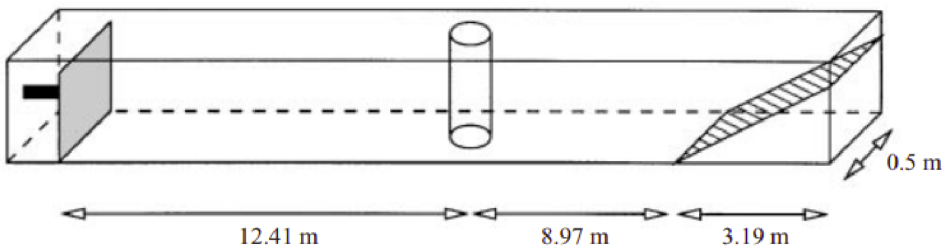


Figure 10: Wave tank configuration with a vertical cylinder placed 12.41 m from the wavemaker in constant depth $h_\mathrm{m} = 0.60$ m [10]

11.2 Geometric Similarity and Model Selection

To achieve accurate upscaling, strict geometric similarity must be preserved between the model and the prototype. The target prototype is a monopile with radius $R_p = 2$ m situated in a water depth of $h_p = 40$ m. The primary non-dimensional geometric parameter is the depth-to-radius ratio, calculated as $h_p/R_p = 20$.

Two model cylinders were available for testing: one with radius $R_m = 0.03$ m and another with $R_m = 0.04$ m. Evaluating the geometric ratio for the smaller cylinder yields $h_m/R_m = 20$, which matches the prototype exactly. Conversely, the larger cylinder yields a ratio of $h_m/R_m = 15$, failing to reproduce the full-scale geometry. Consequently, only the data obtained from the $R_m = 0.03$ m cylinder is used for upscaling to the specified prototype.

11.3 Froude Scaling Framework

Dynamic similarity for gravity-driven surface waves is achieved by equating the Froude number ($\mathrm{Fr} = U/\sqrt{gL}$) between model and prototype. The geometric length scale factor λ_ℓ is defined as the ratio of the water depths:

$$\lambda_\ell = \frac{h_p}{h_m} = \frac{40}{0.60} \approx 66.7 \tag{1}$$

Based on this factor, the scaling laws for the relevant physical quantities are derived. The time scale follows $T_p = \sqrt{\lambda_\ell} T_m$. Kinematically, particle velocities scale as $U_p = \sqrt{\lambda_\ell} U_m$.

The characteristic acceleration $\dot{u}$ scales as amplitude times frequency squared ($a\omega^2$). Since $a_p = \lambda_\ell a_m$ and $\omega_p = \lambda_\ell^{-0.5}\omega_m$, the acceleration scaling factor is $\lambda_\ell \cdot (\lambda_\ell^{-0.5})^2 = 1$. Thus, accelerations are scale-invariant between model and prototype. Assuming inertia dominance where the force F is proportional to fluid density, displaced volume, and acceleration ($F \propto \rho V \dot{u}$), the force scales with the displaced volume (L^3) alone. This results in the force scaling relationship:

$$F_p = \lambda_\ell^3 F_m \approx (66.7)^3 F_m \approx 2.96 \times 10^5 F_m \tag{2}$$

11.4 Theoretical Benchmark: MacCamy-Fuchs

To validate the experimental scaling, the theoretical inline force is calculated using MacCamy-Fuchs diffraction theory [11]. This formulation extends the Morison equation to account for the diffraction of waves by a large vertical cylinder. For a bottom-mounted cylinder in the long-wave regime, the force amplitude F_0 is obtained by integrating the inertia force density over the water depth $z \in [-h, 0]$:

$$F_0 = \int_{-h}^{0} \rho C_M \pi R^2 \cdot \frac{\partial u}{\partial t}\, dz \tag{3}$$

The linear wave acceleration profile is given by $\dot{u} = \omega^2 a \frac{\cosh(k(z+h))}{\sinh(kh)}$. Substituting this into the integral and utilizing the identity $\int \cosh(k(z+h))dz = \frac{1}{k}\sinh(k(z+h))$, the integration over the depth yields:

$$F_0 = \rho C_M \pi R_p^2 \frac{\omega_p^2 a_p}{k_p} \tag{4}$$

where C_M is the effective inertia coefficient. In the long-wave diffraction limit ($kR \ll 1$), this coefficient approaches $C_M \approx 2.0$ [11].

12 Results

12.1 Wave Regime and Slenderness Analysis

Before scaling, the hydrodynamic regime of the experiment was characterized to ensure the applicability of linear theory and the MacCamy-Fuchs formulation [11]. The analysis focuses on the test case with frequency $f_m = 1.425$ Hz.

Solving the linear dispersion relation $\omega^2 = gk\tanh(kh)$ for the model depth $h_m = 0.60$ m yields a wavenumber $k_m \approx 8.17$ m^{-1} and a wavelength $\lambda_m \approx 0.77$ m. The resulting relative depth $h_m/\lambda_m \approx 0.78$ confirms that the experiment was conducted strictly within the deep-water regime ($h/\lambda > 0.5$).

The non-dimensional cylinder size parameter kR was evaluated to determine the relative importance of diffraction. For the model case, $k_m R_m \approx 0.245$, which corresponds to a diameter-to-wavelength ratio of $D/\lambda \approx 0.08$. This places the structure well within the slender body regime ($D/\lambda < 0.2$), where the response is dominated by inertia. However, because kR is non-negligible, the flow is characterized as long-wave diffraction. Consequently, MacCamy-Fuchs theory is applicable with an effective inertia coefficient of $C_M \approx 2.0$.

12.2 Prototype Wave Conditions

The Froude scaling laws derived in the methodology are applied to the specific model wave parameters. The model period $T_m \approx 0.702$ s scales to a prototype period of $T_p \approx 5.73$ s. Solving the dispersion relation for this period at prototype depth (h_p = 40 m) yields a wavelength $\lambda_p \approx 51.3$ m, confirming that the deep water regime is preserved at full scale.

Regarding wave amplitude, the model wave steepness was fixed at $k_m a_m = 0.10$. This yields a model amplitude of $a_m \approx 0.0122$ m. Scaling this by λ_ℓ results in a prototype amplitude of $a_p \approx 0.816$ m, corresponding to a regular wave height of $H_p \approx 1.63$ m.

12.3 Force Estimation Comparison

The concurrent time series of surface elevation and inline force for the analyzed case is shown in Figure 11.

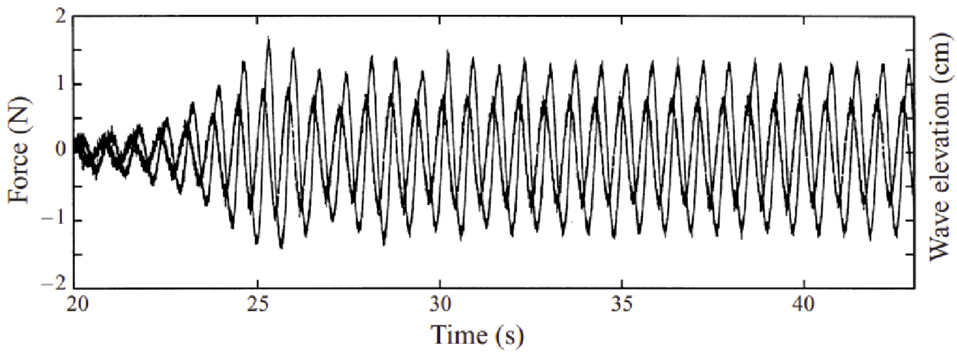


Figure 11: Concurrent time series of surface elevation and inline force ($f_m = 1.425$ Hz) [10].

The prototype force is estimated using the experimentally derived scaling factor from Equation 2, which indicates that $F_p(t) \approx 2.96 \times 10^5 \cdot F_m(t)$. This implies that the measured model forces translate to prototype forces in the order of 300 kN.

This experimental estimate is compared against the theoretical calculation using the MacCamy-Fuchs formulation. With prototype inputs of $\rho = 1025$ kg/m^3, $R_p = 2$ m, $\omega_p = 1.10$ rad/s, and $k_p = 0.123$ m^{-1}, the theoretical force amplitude is calculated as $F_{0,theory} \approx 2.1 \times 10^5$ N.

13 Discussion

The analysis confirms that the selected experimental setup satisfies the strict geometric similarity requirements necessary for valid Froude upscaling to the specified prototype. The rejection of the larger cylinder ($R_m = 0.04$ m) was essential, as it yielded a depth-to-radius ratio of 15, deviating from the prototype ratio of 20 maintained by the selected $R_m = 0.03$ m cylinder.

The hydrodynamic characterization yielded a diameter-to-wavelength ratio of approximately 0.08. While this places the system in the inertia-dominated regime, the calculated slenderness parameter $kR \approx 0.25$ indicates that the structure lies in the transition zone where wave scattering becomes relevant. This justifies the use of MacCamy-Fuchs diffraction theory [11] over a simple constant-coefficient Morison approach, which does not account for the disturbance of the incident wave field by the structure's physical cross-section.

However, a quantitative comparison reveals a substantial discrepancy between the theoretical benchmark and the upscaled experimental loads. The theoretical force amplitude derived from linear diffraction theory is $F_{theor} \approx 2.1 \times 10^5$ N, whereas the Froude-scaled experimental force approaches $F_{exp} \approx 3.0 \times 10^5$ N. This represents a deviation of approximately 43%.

This discrepancy likely arises from the limitations of the theoretical formulation. MacCamy-Fuchs theory assumes inviscid, irrotational potential flow. In reality, even in inertia-dominated regimes, viscous effects can induce flow separation and skin friction, adding a drag component that the potential theory neglects. Furthermore, the experimental waves, though intended to be linear, may exhibit weak non-linearities (Stokes 2^{nd} order effects) in the shallow finite-depth tank, resulting in higher crest particle velocities and accelerations than predicted by linear theory.

Consequently, while the Froude scaling is geometrically valid, the linear theoretical benchmark should be viewed as a conservative lower bound. The higher loads observed in the experiment suggest that design limits for the Bretagne Sud 1 monopile must account for viscous drag contributions and potential wave non-linearities, rather than relying solely on idealized potential flow diffraction theory.

Part IV
Conclusion

14 Conclusion

A multi-scale characterization of the wave climate and associated structural loads was conducted for the Bretagne Sud 1 floating wind project. The study integrated remote sensing, numerical propagation modeling, and experimental scaling to link offshore hind-cast statistics to site-specific design constraints.

Satellite analysis of Sentinel-2 imagery verified the validity of the ResourceCode hind-cast data. Observed wave properties aligned with theoretical dispersion characteristics within statistical uncertainty, while qualitative assessment confirmed the presence of bathymetric refraction in the nearshore region.

Numerical ray-tracing simulations demonstrated that wave propagation across the continental shelf is strongly frequency-dependent. While baseline conditions ($T \approx 14$ s) exhibited diffuse propagation, extreme long-period swells ($T_p > 19$ s) were subject to significant refraction. The bathymetric interaction between the mid-shelf ridge and the adjacent submarine depression resulted in a localized energy flux amplification of approximately 1.4 at the project site. Consequently, the local significant wave height for the 50-year return period was estimated at 22.4 m, exceeding values derived from offshore statistics alone.

Finally, the hydrodynamic analysis of a prototype monopile identified the flow regime as inertia-dominated with non-negligible diffraction effects ($kR \approx 0.25$). However, contrary to initial expectations, load estimates derived from Froude-scaled laboratory measurements exceeded analytical predictions based on MacCamy-Fuchs theory by approximately 43%. This discrepancy suggests that linear diffraction theory acts as a conservative lower bound and fails to capture secondary viscous or non-linear contributions. Therefore, structural design for the site must account not only for the specific bathymetric focusing of long-period spectral components but also for safety factors that accommodate load amplification beyond idealized potential flow predictions.

Declaration of AI Use

Gemini (Alphabet) was used to refine the manuscript's language, formatting, and structure. All scientific interpretations, analyses, and conclusions remain the sole responsibility of the author.

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